UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

International General Certificate of Secondary Education

MARK SCHEME for the May/June 2010 question paper

for the guidance of teachers

0620 CHEMISTRY

0620/21 Paper 21 (Core Theory), maximum raw mark 80

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1 **(a)** methane [1]

(b) methane / propane [1]

(c) ammonia [1]

(d) oxygen [1]

(e) chlorine [1]

(f) ethene [1]

2 **(a)** arrangement: random / far apart OWTTE [1]
motion: random / fast / irregular OWTTE [1]

(b) two paired electrons and two atoms indicated [1]

(c) **(i)** atom of (same) element with different number of neutrons / atoms with same number of protons and different number of neutrons [1]

(ii) number of electrons 1 and 1 [1]
number of neutrons for H-1 = 0 [1]
number of neutrons for H-3 = 2 [1]
number of protons 1 for both [1]

(d) exothermic [1]

(e) **(i)** magnesium>zinc>iron>cobalt [2]
one pair reversed = 1 mark

(ii) calcium chloride; carbon dioxide; water; [3]

3 **(a)** **(i)** reversible / decomposition [1]

(ii) hydrated; water; [2]

(b) **(i)** any two e.g. conducts electricity / conducts heat / sonorous / shiny etc [2]

(ii) forms coloured compounds / forms ions or compounds with variable oxidation state / good catalyst / high melting point OR high boiling point / forms complex ions [2]

(c) reacts with acids / forms a salt and water with acids [1]

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4 **(a)** chloride / Cl^- [1]

(b) K^+ and Br^- (both needed for the mark) [1]

(c) 3.5 (g) [1]

(d) add (nitric acid and) silver nitrate / lead nitrate [1]
yellow ppt [1]

(e) **(i)** I_2 [1]

(ii) brown / yellowish brown [1]
not: grey / black

(iii) bromine is more reactive than iodine OWTTE [1]

(f) 95 [1]

5 **(a)** nitrogen; phosphorus; potassium; [3]

(b) any two of:
plants take up nitrogen / phosphorus / potassium;
nitrogen / phosphorus / potassium needs to be replaced;
to enable better plant growth / greater yield / otherwise plants won't grow as well (idea of increase / more needed) [2]

(c) **(i)** dissolves or idea of dissolving [1]

(ii) titration of acid with alkali / last box ticked [1]

(d) ammonia [1]

(e) **(i)** calcium oxide / lime [1]
allow: calcium hydroxide / limestone / calcium carbonate

(ii) plants grow best at certain pH's / link between pH and plant growth;
farmers want to get best yield; OWTTE [2]

(f) **(i)** 4 [1]

(ii) 15 [1]

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6 **(a)** haematite [1]

(b) **(i)** Any two of:
limestone / coke / air [2]

(ii) iron oxide + carbon → iron + carbon monoxide [2]
1 error = 1 mark

(iii) each arrow or number in the correct position (1 mark each) [4]

(c) ZnS [1]

7 **(a)** boiling point / first box down ticked [1]

(b) fuel oil: fuel for home heating;
kerosene: jet fuel;
lubricating fraction: waxes and polishes;
naphtha: making chemicals; [4]

(c) **(i)** high temperature; catalyst; [2]

(ii) $C_{12}H_{26}$ [1]

(iii) correct structure showing all atoms and bonds [1]

(d) poly(ethene) **allow:** polythene [1]

(e) **(i)** steam [1]

(ii) substance which speeds up rate / speed of reaction [1]

8 **(a)** 1st, 3rd and 4th boxes down ticked (aqueous sodium chloride, copper and graphite) [3]

(b) insulator [1]

(c) **(i)** anode [1]

(ii) negative electrode: zinc [1]
positive electrode: chlorine [1]

(iii) graphite [1]
allow: carbon

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